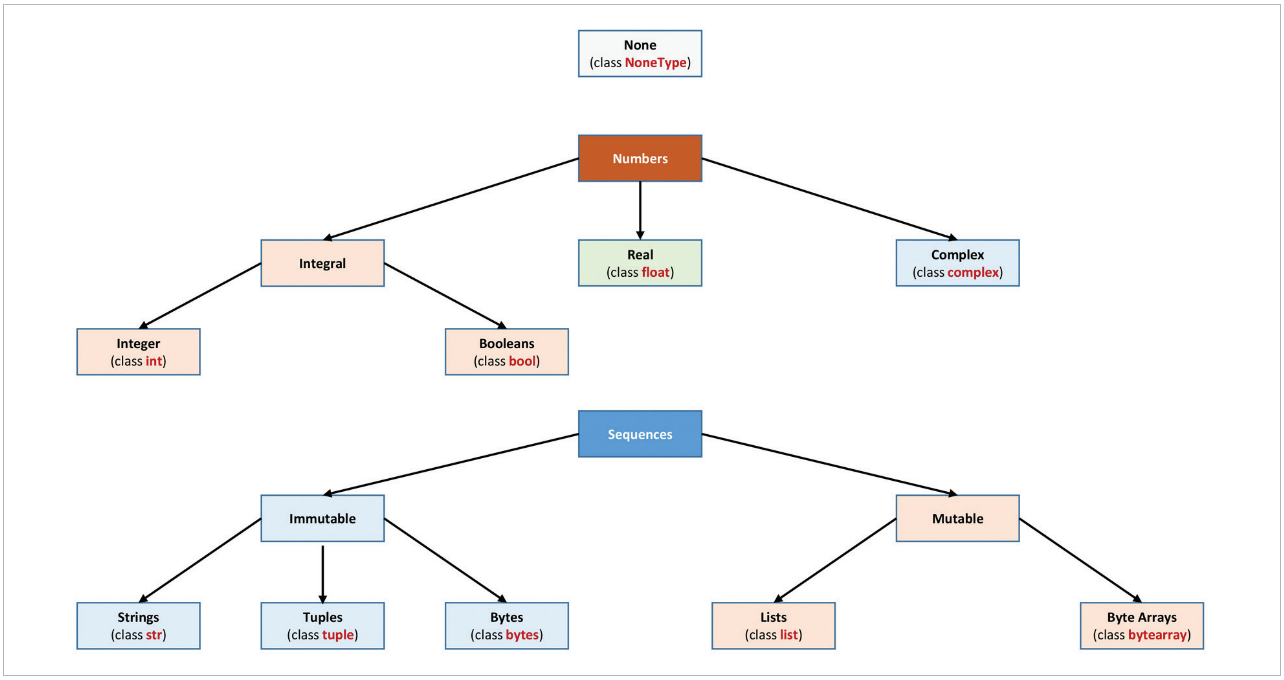

Figure 4: Python data type hierarchy

like the English language and it's easy to comprehend. Another special feature of Python when compared with other object-oriented programming (OOP) languages like C++ and Java is that it's possible to write Python code without making any use of the OOP concept. Python code can be used interpretively. Any Python statement can be interpreted using the interpreter prompt (>>>), and can be executed immediately without compiling the whole program. This is just like how the interpreter works in the BASIC programming language, but such a feature is not available in C/C++ or Java. In Python, variables are not required to be declared explicitly, but in C/C++ and Java, the variables must be declared and their type remains static. For writing various types of code, the ability of the programming language to handle various data types is also important. Python not only supports the primitive data types like Character, Boolean, Integer and Floating Point like C/ C++, Java and R programming languages, but also additional data types like None, Complex Number, Dictionary and Tuple, which makes it more flexible to implement complex algorithms using Python code. Python also supports various mathematical functions with the help of its libraries, which makes it possible to program different types of code for ML.

Cross-compilation is another great feature of Python and any code written in it can be compiled to the C/C++ programming language. This can be done using CPython, which is a reference implementation of Python and can compile Python-like code to C/C++. Another reason for using Python for ML programming is the vast collection of libraries that are designed for the latter. These libraries include Numpy, Scipy, Scikit-learn, Theano, TensorFlow, Keras, PyTorch, Pandas and Matplotlib. Such libraries make it possible to implement ML algorithms with simplicity and convenience. The syntax used in Python is simple and any mathematical statements can be expressed with minimum coding. For example, if you need to implement the Naive Bayes algorithm using Python, it can be done using the code:

```
def bayes_algo(p_a, p_b_a, p_b_not_a):
    not_a = 1 - p_a
    p_b = p_b_a * p_a + p_b_not_a *
not_a
    p_a_b = (p_b_a * p_a) / p_b
    return p_a_b
```

You can see that it's very simple to implement such mathematical formulae using the Python code, without the need to explicitly declare any variable or including any preprocessor/directive like *#include*, in C/C++.

As simple is better than complex, Python can be used for the development of various complex applications with optimised programming code. One empirical study found that Python is more productive than conventional languages, such as C/C++ and Java, for programming problems involving string manipulation or dictionary searches and the memory consumption is also better than Java. That's why many large organisations like Google, Facebook, Amazon, etc, use Python, and it's also helping such companies to grow. END

By: Debojit Acharjee

The author is a software engineer and a writer.